



Course Name:	Discrete Structures	Course Code:	CS1005
Degree Program:	BS(CS/SE/AI)	Semester:	Fall 2023
Exam Duration:	160 Minutes	Total Marks:	120
Paper Date:	Tuesday, December 19, 2023	Obtained Marks:	
Sections:	ALL	No of Page(s):	11
Exam Term & Type:	Final Term I Closed Book	Required Answer Book:	No

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Vetted By

Vetter Signature:

Student Name: _____ Roll No. _____ Section: _____

Instruction/Notes: 1. Attempt all questions on the question-book and in the given order.

<Subjective>

	Q1 (objective)	Q2	Q3	Q4	Q5	Q6	Total
Total Marks	20	40	25	20	20	15	140
Obtained Marks							

Question 2: CLO-3

[40]

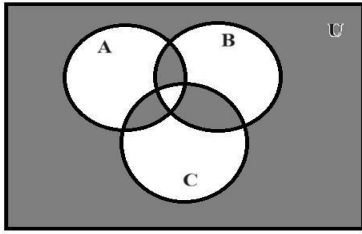
(A neat and clean Venn diagram is important. Cutting and overwriting can directly affect your marks. Use the answer sheet for rough work, perform all calculations, and write a neat and clean answer in the provided space.)

Part (I): Venn Diagram

(10*2=20)

A. A Venn diagram is a graphical representation of a set. Identify the sets corresponding to the following Venn diagrams. (5*2=10)

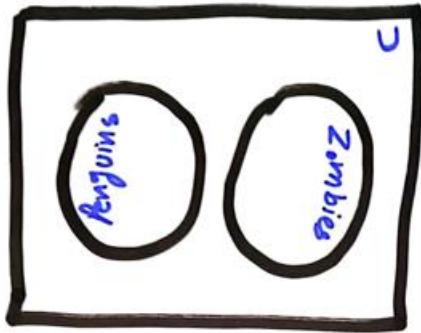
	$A \cap B$
	$(A \cap B) \cup (B \cap C)$



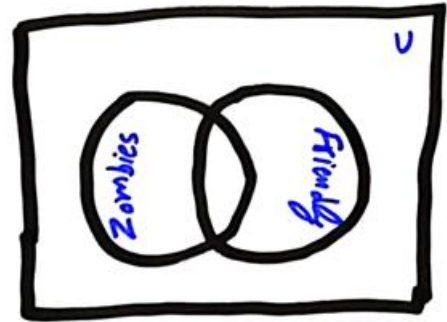
$$\sim(A \cup B \cup C) \cup ((A \cap B) \cup (B \cap C) \cup (A \cap C) - (A \cap B \cap C))$$

B. Draw Venn Diagrams for each of the following statements (iii and iv are pair statements): (2.5*4=10)

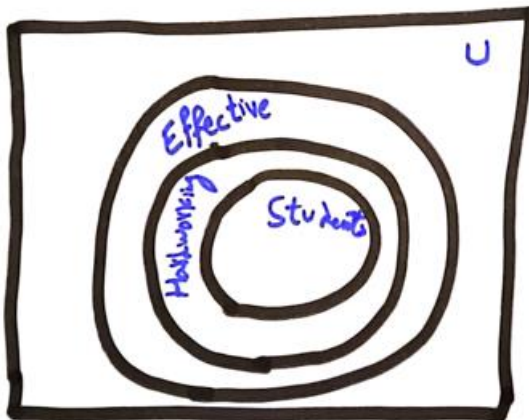
i. No penguins are zombies.



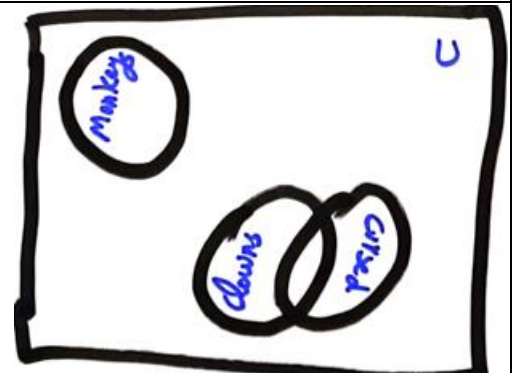
ii. Some zombies are friendly.



iii. All students are hard-working. All hard-working people are effective.



iv. No monkeys are cursed. Some clowns are cursed.



Part (II): Relations

10*2=20

A. If a set $P = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 12\}$. (Tick (✓) if the relation exists and Cross (×) if it does not.)

prove that:

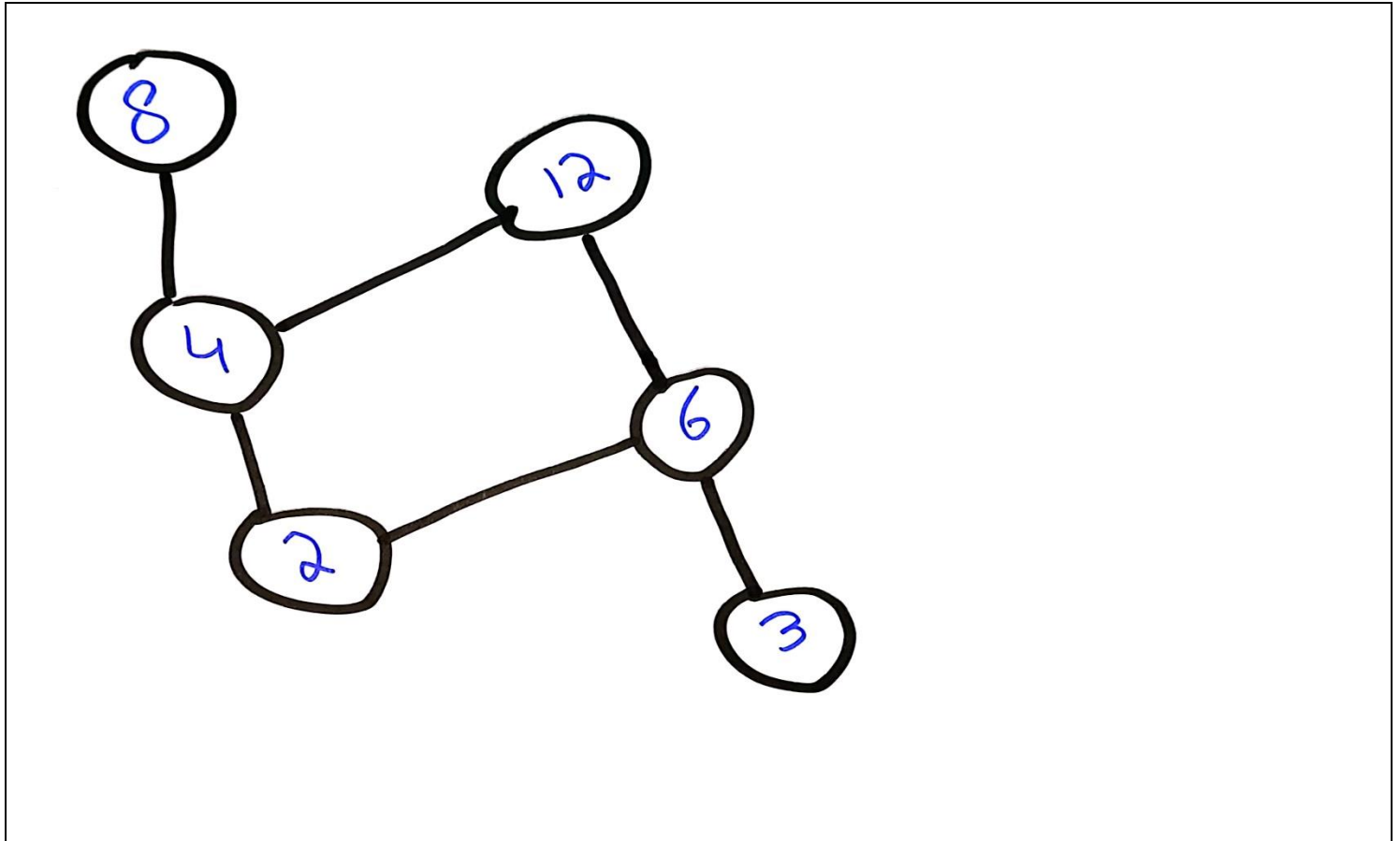
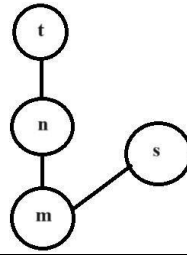
(5*2=10)

Relations	Reflexive	Symmetric	Antisymmetric	Transitive
$R_1 = \{(x, y) x \text{ is divisible by } y\}$	✓	×	✓	✓
$R_2 = \{(x, y) 2x + y \leq 15\}$	×	×	×	×

B. From the relations given previously, choose one relation that is **reflexive**, **antisymmetric**, and **transitive**. Draw the diagram for the relation you chose by following the steps below: (10*1=10)

i. List the elements of the relation.

- ii. Cross out all the reflexive pairs (a, a) in the list.
- iii. Remove all transitive pairs in the list (i.e. if pairs (a, b) and (b, c) exists, cross out the **Transitivity** pair (a, c)).
- iv. Rewrite the elements that are not crossed out.
- v. For every pair (a, b) in final list, draw a line that from point a to point b, where point a is below point b. For example, if your final list contains {(m, n), (n, t), (m, s)}, your drawing will look like the figure below:



Question 3: CLO-3

[25]

(Part-I): The diagram shows the Tarski world in which Triangle(x), Circle(x), and Square(x) mean “x is a triangle,” “x is a circle,” and “x is a square”; Blue(x), Gray(x), and Black(x) mean “x is blue,” “x is gray,” and “x is black”; let RightOf(x, y), Above(x, y), and SameColorAs(x, y) mean “x is to the right of y,” “x is above y,” and “x has the same color as y” for the common domain D of all variables be the set of all the objects in the Tarski world. Convert the given statement in form of formal, logical notation of predicates and also write a negation for each of the given statements. (Note: (Please ensure that you mention the name of law you apply. Failure to do so will result in a deduction of marks for the question). [10]

- a. There is a square x such that x is black. [5]

Statement: $\exists x(\text{Square}(x) \wedge \text{Black}(x))$.

Negation: $\sim(\exists x(\text{Square}(x) \wedge \text{Black}(x)))$

$\equiv \forall x \sim (\text{Square}(x) \wedge \text{Black}(x))$ by the law for negating a $\exists$ statement

$\equiv \forall x(\sim \text{Square}(x) \vee \sim \text{Black}(x))$ by De Morgan's law

- b. For all circles x , there is a square y such that x and y have the same color. [5]

Statement: $\forall x(\text{Circle}(x) \rightarrow \exists y(\text{Square}(y) \wedge \text{SameColor}(x, y)))$.

Negation: $\sim(\forall x(\text{Circle}(x) \rightarrow \exists y(\text{Square}(y) \wedge \text{SameColor}(x, y))))$

$\equiv \exists x \sim (\text{Circle}(x) \rightarrow \exists y(\text{Square}(y) \wedge \text{SameColor}(x, y)))$ by the law for negating a $\forall$ statement

$\equiv \exists x(\text{Circle}(x) \wedge \sim(\exists y(\text{Square}(y) \wedge \text{SameColor}(x, y))))$ by the law for negating an if-then statement

(Part-II): Rewrite the following formal statements in a variety of equivalent but more informal ways. Do not use the symbol $\forall$ or $\exists$. [5]

- a. $\forall x \in \mathbb{R}, x^2 \geq 0$

All real numbers have nonnegative squares.

Or: Every real number has a nonnegative square.

Or: Any real number has a nonnegative square.

Or: The square of each real number is nonnegative.

- b. $\exists m \in \mathbb{Z}^+$ such that $m^2 = m$

There is a positive integer whose square is equal to itself.

Or: We can find at least one positive integer equal to its own square.

Or: Some positive integer equals its own square.

Or: Some positive integers equal their own square

Part-III: if a divides b and b divides c, then a divides c, where a, b, and c belong to integers. By considering the given statement. You are required to Prove it by Direct Proof method. [10]

Assume a, b, and c are integers, such that a divides b, and b divides c.

Then there exist integer r and s, such that,

$$b = a*r \text{ and } c = b*s$$

By substitution

$$c = (a*r)*s$$

$$c = a*(r*s)$$

Let $k=r*s$, then k is an integer, since it is a product of integers, and therefore

$$c = a*k$$

Thus a divides c.

Question 4: CLO-2

[20]

The Genetics Research Institute is conducting a study on a rare genetic disease, Disease-X, which involves three key genetic markers: Marker A (MA), Marker B (MB), and Marker C (MC). The presence of each marker is associated with the probability of an individual developing the disease. The success probabilities for each marker are as follows:

- Probability of having Marker A (MA): $P(MA)=0.6$
- Probability of having Marker B given having Marker A ($MB|MA$): $P(MB|MA)=0.8$
- Probability of having Marker C given having Marker B ($MC|MB$): $P(MC|MB)=0.7$

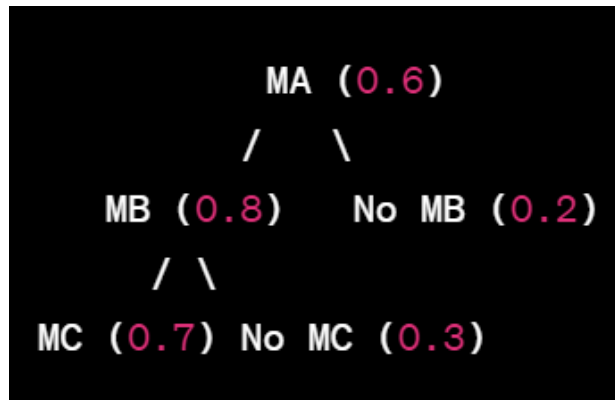
Consider the following genetic scenario:

A family is participating in the study. The probability that at least one family member has Marker A is $P(\text{At least one has MA})=0.9$. If at least one family member has Marker A, the probability that at least one family member has Marker B is $P(\text{At least one has MB}|\text{At least one has MA})=0.75$.

The probability of having Disease-X is $P(\text{Disease-X})=0.05$. Given that a person has Disease-X, the probabilities of having Markers A, B, and C are $P(MA|\text{Disease-X})=0.9$, $P(MB|\text{Disease-X})=0.8$, and $P(MC|\text{Disease-X})=0.7$, respectively.

Answer the following questions:

- a. Develop a probability tree diagram representing the genetic scenario based on the given information. [5]



b. Calculate the overall probability of the following events: [2*3=6]

- Event A: At least one family member has Marker A.

- Event A: At least one family member has Marker A.

$$P(\text{At least one has MA}) = 0.9$$

- Event B: At least one family member has Marker B.

$$\begin{aligned}
 P(\text{At least one has MB}) &= P(\text{At least one has MA}) \times \\
 &P(\text{At least one has MB} | \text{At least one has MA}) \\
 P(\text{At least one has MB}) &= 0.9 \times 0.75 = 0.675
 \end{aligned}$$

- Event C: At least one family member has Marker C.

$$\begin{aligned}
 P(\text{At least one has MC}) &= P(\text{At least one has MB}) \times \\
 &P(\text{At least one has MC} | \text{At least one has MB}) \\
 P(\text{At least one has MC}) &= 0.675 \times 0.7 = 0.4725
 \end{aligned}$$

c. Given that at least one family member has Marker A, calculate the probability of at least one family member having Marker B. Given that at least one family member has Marker B, calculate the probability of at least one family member having Marker C. [4]

- Given that at least one family member has Marker A:

$$P(\text{At least one has MB} | \text{At least one has MA}) = 0.75$$
- Given that at least one family member has Marker B:

$$P(\text{At least one has MC} | \text{At least one has MB}) = 0.7$$

- d. Calculate the probability that a person from the family has Disease-X given that at least one family member has Marker A. Use Bayes' Theorem for this calculation. [5]

Bayes' Theorem is given by:

$$P(\text{Disease-X} | \text{At least one has MA}) = \frac{P(\text{At least one has MA} | \text{Disease-X}) \times P(\text{Disease-X})}{P(\text{At least one has MA})}$$

1. $P(\text{At least one has MA} | \text{Disease-X})$ - The probability that at least one family member has Marker A given that they have Disease-X.
 - Given: $P(\text{At least one has MA} | \text{Disease-X}) = 0.9$
2. $P(\text{Disease-X})$ - The overall probability of having Disease-X.
 - Given: $P(\text{Disease-X}) = 0.05$
3. $P(\text{At least one has MA})$ - The probability that at least one family member has Marker A.
 - Given: $P(\text{At least one has MA}) = 0.9$

Now, we can substitute these values into Bayes' Theorem:

$$P(\text{Disease-X} | \text{At least one has MA}) = \frac{0.9 \times 0.05}{0.9}$$

$$P(\text{Disease-X} | \text{At least one has MA}) = 0.05$$

Question 5: CLO-2

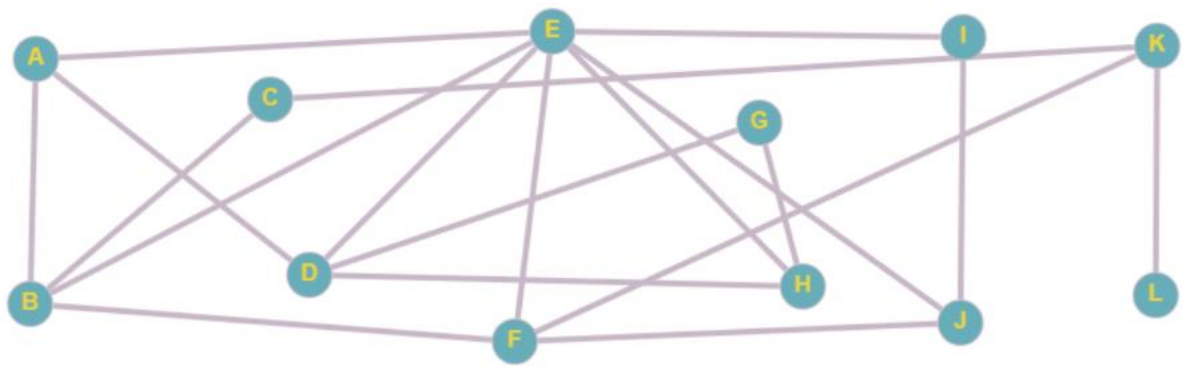
[20]

In an urban setting along a river, a network of bridges connects various districts, creating a complex layout: The layout is given in the following representation where set 'V' represents a set of Vertices and set 'E' represents a set of edges present in graph.

$$V = \{A, B, C, D, E, F, G, H, I, J, K, L\}$$

$$E = \{(A, B), (A, D), (A, E), (B, C), (B, E), (B, F), (C, K), (D, E), (D, G), (D, H), (E, H), (E, F), (E, I), (E, J), (F, J), (F, K), (G, H), (I, J), (K, L)\}$$

- Represent the bridge network within the urban area as a graph, where districts are vertices and bridges are edges. Clearly depict vertices and edges based on the provided layout. [5]



- ii. Investigate the possibility of an Euler Circuit in this graph. If an Euler Circuit exists, illustrate the circuit for the given graph. If not, provide reasoning explaining why an Euler Circuit cannot be formed. [5]

Answer:

Euler Circuit **does not** exist because numerous nodes (A(weight:3), E(weight:7), H(weight:3), J(weight:3), K(weight:3)) exist having odd degree of vertices. For a Euler Circuit to exist, every vertex in the graph must have an even degree.

- iii. Imagine initiating the journey from vertex A and exploring the city by walking across each bridge exactly once, aiming to return to the starting district. Discuss the plausibility of such a path within this graph. [5]

Answer:

No Such possibility exists because of the graph having odd degree of numerous vertices. To be more specific, while traversing through vertex 'L' at least one bridge will be repeated and hence an Euler circuit would not be formed.

Graph Adjustment Proposition:

Recommend necessary modifications (if any) to the graph's structure to enable the potential traversal of the city bridges while ensuring each bridge is crossed exactly once, culminating in a return to the starting district. (5 Marks)

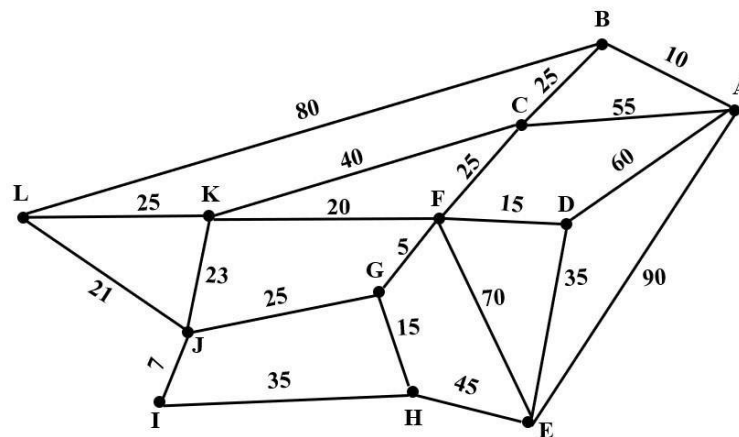
- iv. Recommend necessary modifications (if any) to the graph's structure to enable the potential traversal of the city bridges while ensuring each bridge is crossed exactly once, culminating in a return to the starting district. [5]

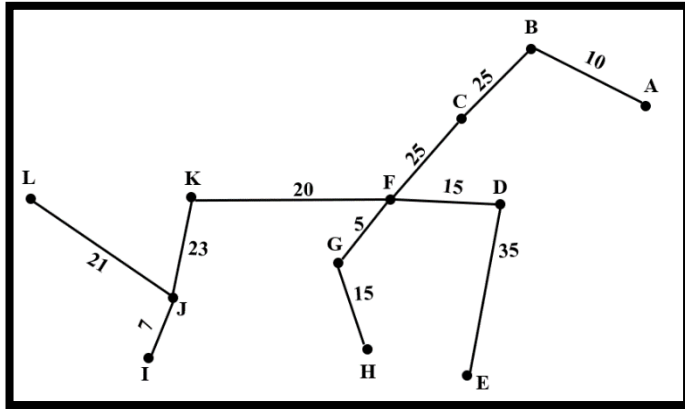
Answer:

Changing the odd degrees of vertices for every vertex in the graph to an even degree would fix the problem and make the traversals possible. Notice that destroying any vertex is not an acceptable solution while making more is acceptable. Students would come up with solutions containing graphs. The graphs or subgraphs conforming to the properties of Euler Circuit are an acceptable solution.

Question 6: CLO-2**[15]**

- a. The roads represented by this graph are all unpaved. The lengths of the roads between pairs of towns (A to L) are represented by edge weights. Which roads should be paved so that there is a path of paved roads between each pair of towns so that a minimum road length is paved? Use **Prim's algorithm** to find a minimum spanning tree for the given weighted graph. Start from town A and also maintain sequence of all edges along with its vertices in tabular form. **[8]**

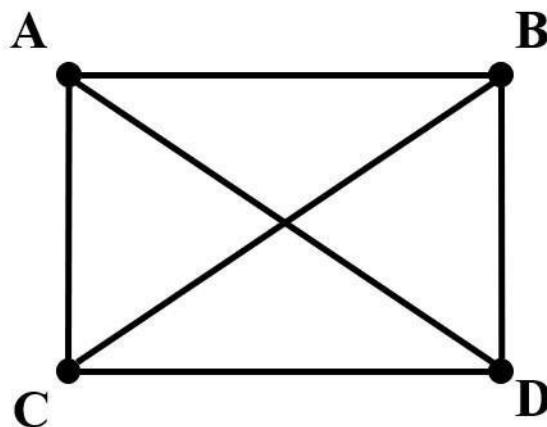




Choice	Edge	Weight
1	{A, B}	10
2	{B, C}	25
3	{C, F}	25
4	{F, G}	5
5 OR 6	{F, D}	15
5 OR 6	{G, H}	15
7	{F, k}	20
8	{K, J}	23
9	{J, I}	7
10	{J, L}	21
11	{D, E}	35

b. Consider the following graph G and answer the following questions: [7]

- I. Use KIRCHHOFF'S THEOREM to find the number of spanning trees. [4]
- II. Which method is best to find the number of spanning trees easily for this graph. [2]
- III. How many edges need to remove for single spanning tree in this graph. Write proper formula. [1]



I. By KIRCHHOFF'S THEOREM:

$$A(G) = \begin{array}{c} \begin{array}{cccc} & a & b & c & d \\ \begin{array}{c} a \\ b \\ c \\ d \end{array} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \end{array}$$

$$M = \begin{array}{c} \begin{array}{cccc} & a & b & c & d \\ \begin{array}{c} a \\ b \\ c \\ d \end{array} & \begin{bmatrix} 3 & -1 & -1 & -1 \\ -1 & 3 & -1 & -1 \\ -1 & -1 & 3 & -1 \\ -1 & -1 & -1 & 3 \end{bmatrix} \end{array}$$

$$M = \begin{vmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{vmatrix}$$

$$= 3 \begin{vmatrix} 3 & -1 \\ -1 & 3 \end{vmatrix} - (-1) \begin{vmatrix} -1 & -1 \\ -1 & 3 \end{vmatrix} + (-1) \begin{vmatrix} -1 & 3 \\ -1 & -1 \end{vmatrix}$$

$$= 3(9-1) - (-1)(-3-1) + (-1)(1+3) = 16$$

II. Best method:

Answer: $n^{n-2} = 4^{4-2} = 4^2 = 16$ because this is complete graph.

III. How many edges need to be removed?

Answer: $e - v + 1 = 6 - 4 + 1 = 3$

Objective:

Question	Correct Answer
1	B) $\neg p \vee \neg q$
2	B) 1
3	A) $p \wedge \neg q$
4	A) Tautology
5	D) $(p \vee q) \wedge \neg (p \wedge q)$
6	A) $p \wedge \neg q$
7	B) $\neg (p \vee q) = \neg p \wedge \neg q$
8	B) $\neg q \rightarrow \neg p$
9	B) $\neg p \vee \neg q$
10	A) True
11	C) If it is sunny, then I'll go for a walk. I'm going for a walk; therefore, it is sunny.
12	C) Simplification
13	D) Law of contradiction
14	A) Some cars are not vehicles.
15	A) Distributive law
16	B) If the streets are not wet, then it is not raining.
17	B) Commutative law
18	C) If it is not hot, then I won't go swimming.
19	A) Distributive law
20	C) No dogs are mammals.

GOOD LUCK 😊